

GEMEA: An Open Knowledge Graph over 26M+ German Cultural Heritage Objects

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Abstract. GEMEA is the first open knowledge graph over the German Digital Library (DDB) corpus, making 26.8 million digitized objects queryable via SPARQL, browsable as Linked Data, and accessible to AI agents through a Model Context Protocol (MCP) interface. The MOCHO alignment layer maps DDB-EDM metadata to RDA and domain-specific ontologies through corpus-driven dispatch, with GND authority linking for literary and musical works. GEMEA supports research on KG construction, ontology evaluation, and LLM-grounded cultural heritage retrieval.

Keywords: knowledge graphs · Linked Open Data · digital humanities · ontology alignment · research infrastructure · Model Context Protocol.

1 Introduction

The German Digital Library (Deutsche Digitale Bibliothek, DDB)⁴ aggregates over 65M objects⁵ from German memory institutions — archives, libraries, museums, media libraries, and monument preservation offices, referred to hereafter as *sectors* — all described using the Europeana Data Model (EDM)⁶. The corpus has no public SPARQL endpoint, no Linked Data browser, and no machine-readable interface for KG-based research.

Converting DDB-EDM records to a queryable knowledge graph (KG) is complicated by sector-specific metadata conventions that diverge across institutional traditions. Each sector applies locally-specific field interpretations and carries cataloging practices into every record; no cross-sector authority governs how EDM properties are instantiated. This variation is invisible to schema-level or structure-based alignment tools; corpus-based analysis is therefore a precondition for reliable automated alignment.

Three contributions are described in this paper:

⁴ DDB, <https://www.deutsche-digitale-bibliothek.de>

⁵ of which 27M are digitized

⁶ EDM, <https://pro.europeana.eu/page/edm-documentation>

1. **Resource.** GEMEA /gə'mera/ (**German Memory Atlas**), a publicly queryable KG of 26M German cultural heritage objects, served via a QLever SPARQL endpoint [1] and browsable through a SHMARQL[14] Linked Data interface⁷ — the first open, SPARQL-accessible KG over the DDB corpus.
2. **Ontology alignment Layer.** A MOCHO⁸-based[15] /'mo.tʃo/ alignment layer mapping DDB-EDM metadata to RDA/FRBR-compatible representations, demonstrated on a representative subset, with alignment gaps documented as findings rather than omitted.
3. **Agentic access layer.** An MCP/MCPO interface on the QLever endpoint, enabling LLM-based agents to issue SPARQL queries, retrieve entity descriptions, and integrate GEMEA into multi-step reasoning workflows.

As a KG derived from in-use institutional data, GEMEA is intended as a research testbed; Section 5 details supported use cases and downstream research scenarios. Data is licensed CC BY 4.0; the SPARQL endpoint, Linked Data browser, and source code are available at <https://gemea.ise.fiz-karlsruhe.de>.

Paper structure. Sections 2–7 cover related work, resource description, quality, usage, impact, and sustainability in turn.

2 Related Work

This section reviews prior work on CH KGs, ontology alignment approaches, and interfaces for querying and exploring large-scale Linked Data. **CH knowledge graphs and LOD initiatives.** Europeana LOD [7] is the largest CH Linked Open Data initiative, exposing EDM-compliant metadata for tens of millions of objects aggregated from across Europe; its SPARQL endpoint has been discontinued and data is distributed primarily as bulk download, with no interactive graph browser. The German National Library’s Integrated Authority File or *Gemeinsame Normdatei* (GND) is a major authority dataset maintained by the German National Library (DNB). The GND is accessible as a bulk download⁹, via the lobid REST API¹⁰ and via a QLever-hosted SPARQL endpoint,¹¹ but none provides a dedicated exploratory Linked Data browser.

Domain-specific CH knowledge graphs. ArCo [4] is an Italian CH KG modeled several domain ontologies including CIDOC-CRM¹², a conceptual reference model for structuring CH information, and interlinked with DBpedia¹³ and the Getty Vocabularies¹⁴, covering around 820K cultural objects; it provides

⁷ SHMARQL, <https://github.com/epoz/shmarql>

⁸ Middle Ontology for Cultural Heritage Objects

⁹ GND Open Data, <https://data.dnb.de/opendata/>

¹⁰ lobid REST API, <https://lobid.org/gnd/api>

¹¹ GND SPARQL Endpoint, <https://sparql.dnb.de/dnbgnd>

¹² CIDOC CRM, <http://www.cidoc-crm.org/rdfs/cidoc-crm#>

¹³ <https://www.dbpedia.org/>

¹⁴ <https://www.getty.edu/research/tools/vocabularies/>

a SPARQL endpoint but no graph browser. PHAROS¹⁵ is an international consortium of photo archives exposing millions of art-historical photographs via a public SPARQL endpoint and discovery interface; coverage is limited to visual art photography.

ArtKB [2]. ArtKB is a multimodal knowledge base for art objects using the CACAO ontology (a CIDOC-CRM extension), supporting Text2SPARQL and neural image retrieval over 307K curated objects from Wikidata; it exposes no public SPARQL endpoint, includes no Linked Data browser, and does not document self-hosting.

Ontology alignment in CH KGs. Ontology alignment has been explored in CH KGs, particularly for mapping domain-specific metadata schemas to conceptual reference models such as FRBR [6]. Existing approaches focus on linking institutional cataloging schemas to shared ontologies, often via manual or semi-automatic mappings. MOCHO [15] extends this line of work by mapping `edm:ProvidedCHO`¹⁶ to FRBR [6] levels via GND Werk links, aligning DDB-EDM to domain- and modality-specific ontologies. GEMEA extends MOCHO-based alignment to the full DDB corpus.

SPARQL endpoints and KG browsers. Public SPARQL endpoints such as the Wikidata Query Service¹⁷ provide domain-agnostic, structured query access to KGs, but typically require users to formulate SPARQL queries and do not support interactive exploration. In contrast, generic LOD browsers provide a mechanism for navigable exploration of RDF resources via dereferencing and faceted or graph-based interfaces, but do not support complex query-driven or collection-level analysis. SHMARQL [14] provides a Linked Data browser and SPARQL proxy deployable via Docker over any existing endpoint.

To the best of our knowledge, GEMEA is the first to combine national-scale cross-domain coverage with a public SPARQL endpoint, an interactive Linked Data browser, and self-hostable 352deployment (Table 1). The MCP/MCPO agentic access layer is not documented in prior cultural heritage KG work.

3 Resource Description

GEMEA is a knowledge graph over approximately 26.8 million objects from the German Digital Library (DDB), organized into named graphs in QLever and served through SHMARQL¹⁹ as a SPARQL endpoint and Linked Data browser; the dataset is available for download at <https://gemea.ise.fiz-karlsruhe.de/downloads>, and the MCP/MCPO integration is released as self-hosting setup scripts.

¹⁵ <https://artresearch.net/sparql>

¹⁶ All CH objects in EDM are instances of this class.

¹⁷ <https://query.wikidata.org/>

¹⁹ <https://gemea.ise.fiz-karlsruhe.de/shmarql/>

Table 1: Comparison of GEMEA with related cultural heritage knowledge graphs and interfaces. *Scale* = approximate object count; *SPARQL* = public SPARQL endpoint; *Browser* = interactive Linked Data or graph browser; *Self-host* = self-hostable deployment documented.

System	Scale	Ontology	SPARQL	Browser	Self-host
GEMEA (this work)	26.8M	DDB-EDM + MOCHO (RDA/FRBR)	✓	SHMARQL	✓
ArtKB [2]	307K	CACAO (CIDOC-CRM)	–	–	–
PHAROS	>1M	–	✓	artresearch.net	–
Europeana LOD [7]	~50M	EDM	–	–	–
ArCo [4]	820K	CIDOC-CRM	✓	–	–
DDB ¹⁸ native UI	65M	DDB-EDM + Proprietary	–	search only	–

3.1 GEMEA

The source data consists of the digitized subset of the DDB corpus: the DDB aggregates metadata records from more than 500 German cultural heritage institutions across 7 sectors (*Bibliothek, Archiv, Museum, Mediathek, Denkmal, Forschung, Sonstiges*), many of which describe objects that have not been scanned, photographed, or recorded; GEMEA covers only those records for which a digital surrogate exists.

DDB JSON-LD records are converted to N-Quads by a custom transformation pipeline that maps EDM predicates to RDA and mocho (Section 3.2) properties and partitions output into named graphs by content layer. A post-processing phase links literary and musical works to GND authority records through several passes employing NLP models for Information Extraction, adding work-level links to a fourth named graph. The QLever index is organized as four named graphs: **graph/ddbedm** (verbatim EDM passthrough), **graph/mocho** (mocho-aligned triples), **graph/prov** (PROV-O provenance), and **graph/work** (GND work links), allowing queries to target a single content layer without full-corpus scans.

Scale. Digitized DDB objects are retrieved by querying the DDB Search API²⁰, filtering for digitized objects only (**digitalisat:true**); JSON records are fetched per object from the DDB Items API.²¹ During serialisation to RDF triples, malformed IRIs are percent-encoded to produce valid IRI references, and internal DDB identifiers are replaced with **urn:** URNs to ensure global uniqueness of entity URIs across the graph. Rights are not uniform and are queryable directly via **dcterms:rights**; see Section 4.3.

²⁰ <https://api.deutsche-digitale-bibliothek.de/2/search/>

²¹ <https://api.deutsche-digitale-bibliothek.de/2/items/<DDB object ID>>

Table 2: Object and triple counts per sector.

Sector	Objects	Total triples	Named graph triples		
			DDB-EDM	MOCHO	PROV
Library	18,338,116	1,930,178,220	1,040,390,087	356,952,292	532,835,841
Archive	3,456,119	467,439,123	259,483,277	61,795,115	146,160,731
Museum	2,011,841	230,484,127	123,795,378	48,164,578	58,524,171
Media Library	1,709,846	258,176,923	145,905,243	58,871,240	53,400,440
Research	1,165,891	138,118,201	75,842,237	24,633,128	37,642,836
Monument Preserv.	79,393	9,020,751	4,852,432	1,555,297	2,613,022
Others	85,408	9,996,438	5,487,618	1,958,508	2,550,312
Total	26,846,614	3,043,413,783	1,655,756,272	553,930,158	833,727,353

3.2 MOCHO Alignment

EDM is designed for cross-institutional aggregation: all objects share a flat `edm:ProvidedCHO` type, DC/EDM predicates carry no WEMI-level distinction, and agent roles are untyped, making work-level grouping, typed predicate queries, and structured semantic querying over the collection dependent on further alignment. The **MOCHO** ontology [15] addresses this by assigning domain- and mediatype-specific types to `ProvidedCHOs`, routing EDM predicates to typed properties at the correct WEMI level, and grouping co-manifesting records into `mocho:Work` entities where GND authority links are available. The alignment is developed and validated on the *Goethe-Faust corpus* (115,432 records), which serves as the reference implementation corpus before scaling to the full 26.8M-object DDB collection. The full alignment table and per-class statistics are released in the GEMEA GitHub repository²².

Corpus-driven class dispatch. A key finding from corpus analysis is that DDB-EDM property semantics are sector-dependent and not declared in the schema: `dc:type` carries a mixture of free-text terms and controlled-vocabulary in library records, free-text strings in museum records, and structural hierarchy codes in archive records; `ddb:hierarchyType` (htype) is present in archive and library strata but mostly absent in non-TEXT media types across museum, media library, and monument sectors entirely. These differences surface only through corpus measurement. Signal coverage is therefore quantified per sector $\times$ mediatype stratum before any dispatch logic is implemented: Archive text records carry both htype and dc:type in 99.6% of cases; Library text records are htype-only in 31%; Museum, Media Library, and Monument records carry no htype at all. This coverage profile sets dispatch priority per stratum: htype-first for Archive and Library, dc:type-first for the remaining sectors, rather than a uniform fallback chain. Scope boundaries where real EDM encodings deviate from the alignment assumptions are documented in Section 4.

²² <https://github.com/anntanp/gemea/>

Iterative ontology extension. The target ontology is not fixed before transformation begins; instead, corpus analysis drives its extension. Entity types encountered during inspection of the Goethe-Faust corpus (digitized photographs, cartographic objects, immovable cultural monuments) had no adequate representation in MOCHO before the alignment work. New classes (`mocho:ImageWork`, `mocho:ImmovableWork`, `mocho:ImageObject`) are introduced in direct response to patterns surfaced in the corpus: a gap is identified, a class is defined in the ontology, alignment rules are updated, and the transform is re-run. This cycle repeats until alignment coverage is stable across all strata.

Property mapping. Because DDB-EDM properties are DC²³/DCT²⁴-based, the RDA Registry DC-to-RDA mapping²⁵ provides the systematic bridge to typed RDA predicates at the correct WEMI level. RDA serves as the alignment pivot: each assigned class branches to its native domain ontology (general metadata for library records, VRA Core for visual objects, RiC-O for archival materials, Music Ontology for audio), without requiring separate DC-to-domain mappings per ontology. Target predicates are resolved per (`target_class`, `ddb-edm_prop`) pair, with corpus inspection correcting derivation errors in the machine-generated alignment table and resolving predicate choices where the table produces spurious candidates. For contributor roles, the LIDO event type of the `edm:Event` node linked to the `edm:ProvidedCHO` determines the emitted property (author, photographer, publisher), exploiting the event-centric aspect of EDM to capture typed role semantics invisible at the DC level.

3.3 SHMARQL Linked Data Interface

SHMARQL [14] is a lightweight Linked Data publishing layer deployed as a Docker container (ghcr.io/epoz/shmarql). Entity URIs are dereferenceable, RDF is served with content negotiation (HTML, Turtle, JSON-LD), and a built-in SPARQL query editor is included. As a SPARQL-endpoint agnostic browser, SHMARQL is redeployable without modification and provides the primary dereferencing interface for GEMEA's `urn:-minted` entity URIs for global uniqueness, which carry no native HTTP resolution.

For GEMEA, SHMARQL is deployed as a proxy over the QLever endpoint. The deployment was validated on the Goethe-Faust corpus before scaling to GEMEA. The GEMEA SHMARQL instance is available at <https://gemea.ise.fiz-karlsruhe.de/shmarql>. Figure 1 shows an example of the GEMEA SHMARQL interface for interactive SPARQL query and entity-centric graph exploration.

²³ Dublin Core Elements (`dc:`), <http://purl.org/dc/elements/1.1/>

²⁴ DCMI Metadata Terms (`dcterms:`), <http://purl.org/dc/terms/>

²⁵ <https://www.rdaregistry.info/Maps/mapRDA2DCT.html>

The screenshot shows the SHMARQL web interface. At the top is a blue header with the SHMARQL logo and a hamburger menu. Below the header is a 'Prefixes' tab. The main area contains a SPARQL query editor with the following code:

```

1 PREFIX edm: <http://www.europeana.eu/schemas/edm/>
2 PREFIX dc: <http://purl.org/dc/elements/1.1/>
3 SELECT ?cho ?title WHERE {
4   ?agg edm:aggregatedCHO ?cho .
5   ?cho dc:creator <http://d-nb.info/gnd/118540238> ;
6       dc:title ?title .
7 }
8 LIMIT 100

```

Below the query editor, it indicates '100 results in 27ms'. The results are displayed in a table with two columns: '# cho' and 'title'. The table contains three rows of results, each with a URL in the 'cho' column and a title in the 'title' column. Under each URL in the 'cho' column, there are three small buttons labeled 'S', 'P', and 'O'.

#	cho	title
1	http://www.deutsche-digitale-bibliothek.de/item/227H6MUY4Q2VX5RV5YTQHB7QFAHHJSNU S P O	Voyage en Italie ^{zxx}
2	http://www.deutsche-digitale-bibliothek.de/item/22KEGSLBX7XK4OUMHZ7VLFUXUL75IMNZE S P O	Werke in 6 Bden, Bd. 1.. Gedichte [u. a.] ^{zxx}
3	http://www.deutsche-digitale-bibliothek.de/item/22MBKP7UGKMJ3MIGSXD7J2XNYFC6PZZE S P O	Wie herrlich leuchtet mir die Natur : Gedichte und Bilder ^{zxx}

Fig. 1: SHMARQL interface showing the SPARQL query editor and entity view. The [S], [P], and [O] controls under each entity in the CHO column load all triples where that entity appears as subject, predicate, or object, enabling step-by-step graph traversal without writing SPARQL.

3.4 MCP/MCPO Agentic Access Layer

The Model Context Protocol (MCP) ²⁶ is an open standard for exposing tools and data sources to AI agents. `mcp-server-qllever`²⁷ is a third-party MCP server tailored for the QLever engine, exposing SPARQL query execution, schema exploration, entity search, and QLever-specific capabilities (context-sensitive autocompletion, spatial queries, query plan analysis) as MCP tools. GEMEA's QLever endpoint is registered with `mcp-server-qllever`, making the KG queryable to MCP-compatible agents or LLM clients without custom integration code.

²⁶ MCP, <https://modelcontextprotocol.io/docs/getting-started/intro>

²⁷ <https://mcpmarket.com/server/qllever>

An MCPO²⁸ wrapper is also provided, exposing the endpoint as an OpenAPI-compatible HTTP interface for REST-native clients; invocation support depends on the client framework.

Setup scripts for the full stack, including MCP registration, are provided in the GEMEA GitHub repository.²⁹

Example agent interaction. Figure 2 shows an example of multi-step KG reasoning over GEMEA via the MCP interface. An agent is asked: “How many objects in the dataset are attributed to Goethe? Which institutions hold the most of them?” The agent resolves Goethe’s GND identifier (`gnd:118540238`), issues a `dc:creator` query, and groups results by `edm:dataProvider`:

```
PREFIX dc: <http://purl.org/dc/elements/1.1/>
PREFIX edm: <http://www.europeana.eu/schemas/edm/>
SELECT ?provider
      (COUNT(DISTINCT ?cho) AS ?n)
WHERE {
  ?cho dc:creator
    <http://d-nb.info/gnd/118540238> .
  ?agg edm:aggregatedCHO
    ?cho ;
    edm:dataProvider
      ?provider .
  FILTER(isLiteral(?provider))
}
GROUP BY ?provider
ORDER BY DESC(?n)
```

How many objects in the dataset are attributed to Goethe? Which institutions hold the most of them?

- Thought for 1s >
- Bash** Count total objects attributed to Goethe


```
IN curl -s -G "http://localhost:7030/api" \
  --data-urlencode "query=PREFIX dc: <http://purl.org/dc/elements/1.1/>
  PREFIX edm: <http://www.europeana.eu/schemas/edm/>"
OUT {"head":{"vars":["total"]},"results":{"bindings":[{"total":{"datatype":"http://www.w3.org/2001/XMLSchema#integer","value":7788}]}}
```
- Bash** Objects attributed to Goethe grouped by provider


```
IN curl -s -G "http://localhost:7030/api" \
  --data-urlencode "query=PREFIX dc: <http://purl.org/dc/elements/1.1/>
  PREFIX edm: <http://www.europeana.eu/schemas/edm/>"
OUT {"head":{"vars":["provider","count"]},"results":{"bindings":[{"provider":{"datatype":"http://www.w3.org/2001/XMLSchema#string","value":"Deutsche Nationalbibliothek","type":"literal"},"count":{"datatype":"http://www.w3.org/2001/XMLSchema#integer","value":3551,"type":"literal"}}, {"provider":{"datatype":"http://www.w3.org/2001/XMLSchema#string","value":"Bayerische Staatsbibliothek","type":"literal"},"count":{"datatype":"http://www.w3.org/2001/XMLSchema#integer","value":2064,"type":"literal"}}, {"provider":{"datatype":"http://www.w3.org/2001/XMLSchema#string","value":"Landesarchiv Thüringen – Hauptstaatsarchiv Weimar","type":"literal"},"count":{"datatype":"http://www.w3.org/2001/XMLSchema#integer","value":1639,"type":"literal"}}, {"provider":{"datatype":"http://www.w3.org/2001/XMLSchema#string","value":"SLUB Dresden. Musik und AV-Medien","type":"literal"},"count":{"datatype":"http://www.w3.org/2001/XMLSchema#integer","value":87,"type":"literal"}}, {"provider":{"datatype":"http://www.w3.org/2001/XMLSchema#string","value":"Archiv des Heine-Instituts und Schumann-Hauses","type":"literal"},"count":{"datatype":"http://www.w3.org/2001/XMLSchema#integer","value":80,"type":"literal"}}, {"provider":{"datatype":"http://www.w3.org/2001/XMLSchema#string","value":"Herzogin Anna Amalia Bibliothek","type":"literal"},"count":{"datatype":"http://www.w3.org/2001/XMLSchema#integer","value":65,"type":"literal"}}, {"provider":{"datatype":"http://www.w3.org/2001/XMLSchema#string","value":"","type":"literal"},"count":{"datatype":"http://www.w3.org/2001/XMLSchema#integer","value":0,"type":"literal"}}]}}
```
- 7,788 objects** attributed to Goethe (GND **118540238**). Top institutions:

Provider	Objects
Deutsche Nationalbibliothek	3,551
Bayerische Staatsbibliothek	2,064
Landesarchiv Thüringen – Hauptstaatsarchiv Weimar	1,639
SLUB Dresden. Musik und AV-Medien	87
Archiv des Heine-Instituts und Schumann-Hauses	80
Herzogin Anna Amalia Bibliothek	65

The top 3 account for ~94% of Goethe-attributed objects.

Fig. 2: SPARQL query issued by the agent (left) and the Claude Code MCP session (right). The query returns 7,788 objects across 10 institutions; the three largest holders are the Deutsche Nationalbibliothek (3,551), Bayerische Staatsbibliothek (2,064), and Landesarchiv Thüringen – Hauptstaatsarchiv Weimar (1,639).³⁰

²⁸ MCPO, <https://github.com/open-webui/mcpo>

²⁹ <https://github.com/anntanp/gemea/>

³⁰ Demo run on the Goethe-Faust reference corpus [15].

4 Quality and Validation

MOCHO was built with the Ontology Development Kit (ODK)³¹. The ontology requires OWL 2 DL expressivity: it makes use of universal restrictions (`owl:allValuesFrom`), cardinality constraints, complement and disjointness axioms, and property chains, which collectively place it outside the OWL 2 EL profile.

OWL 2 DL consistency is verified by HermiT [12] as part of the ODK build; materialized inferences are exported as RDF triples and loaded into QLever alongside the DDB corpus, so no runtime reasoning is performed by the triplestore. The ontology and its documentation are published through the ODK and Widoco [5] pipeline, producing both the machine-readable artifact and human-readable HTML documentation from a single build step. The EDM-to-MOCHO property mappings are validated by a suite of unit tests that assert correct predicate assignment per (`target_class`, `ddb-edm_prop`) pair against gold-standard triples derived from the Goethe-Faust corpus.

4.1 Completeness

GEMEA is a snapshot exported between late March and early April 2026, covering 97.7% of the digitized DDB corpus at time of writing (Table 2). The remaining gap reflects objects added to the DDB after the snapshot window and a small number of transient API search failures during the export run.

4.2 Scope of Validation

Comprehensive metadata quality evaluation at the scale of GEMEA requires infrastructure for corpus-level querying, inspection, and reproducible analysis. GEMEA is released to provide this infrastructure: the public SPARQL endpoint, Linked Data browser, and downloadable dumps enable systematic downstream evaluation of metadata quality, alignment behavior, and retrieval performance across the corpus.

4.3 Known Limitations

- **Snapshot currency.** The dataset reflects the DDB as of late March to early April 2026. Records added, updated, or deleted after the export window are not reflected; no incremental update mechanism is currently implemented.
- **Partial GND linking.** Werk linking covers literary and musical works but achieves low recall due to inconsistent encoding of `dc:title` values across contributing institutions: corpus analysis of the Goethe-Faust reference corpus shows 71% of titles carry no ISBD punctuation [8], with the dominant signal being the subtitle separator ‘:’ (18%) rather than the statement-of-responsibility separator ‘/’ (2.1%) used for work-title extraction; corpus-wide

³¹ <https://incatools.github.io/ontology-development-kit/>

figures for 9.2M German-language DDB titles show 92.4% without any ISBD marker. Future work will apply ISBD punctuation heuristics and a planned NER model fine-tuned on a DDB subset to improve coverage; authority linking for images of artworks and architecture via Wikidata, Bildindex³², and artresearch.net’s PHAROS infrastructure is also planned.

- **Alignment coverage at scale.** The EDM-to-MOCHO mapping was developed and validated against the Goethe-Faust corpus (115,432 library records). Sector-specific edge cases, particularly in the Library sector at 18M records, may produce fallback or semantically underspecified type assignments.
- **Rights metadata.** Each object retains the `dcterms:rights` value as declared by the contributing institution; rights statements are not normalized, verified, or mapped to a controlled vocabulary.
- **Geographic and temporal sparsity.** Corpus analysis of the Goethe-Faust reference corpus shows that only 45% of objects carry any date value (`dc:date`, `dcterms:date`, or `dcterms:created`) and only 7.1% carry a place value (`dc:coverage` or `dcterms:spatial`).

5 Usage and Reusability

5.1 SPARQL Access

A public read-only SPARQL 1.1 endpoint is available at <https://gemea.ise.fiz-karlsruhe.de/sparql>, backed by QLever. Listings 1–3 illustrate representative query patterns; Listing 3 demonstrates cross-graph querying over `graph/ddbedm` and `graph/mocho`.

Listing 1.1: Objects associated with a given agent (GND URI).

```
PREFIX edm: <http://www.europeana.eu/schemas/edm/>
PREFIX dc: <http://purl.org/dc/elements/1.1/>
SELECT ?cho ?title ?provider WHERE {
  ?cho dc:creator <http://d-nb.info/gnd/118540238> ; # Goethe
      , Johann Wolfgang von
      dc:title ?title .
  ?agg edm:aggregatedCHO ?cho ;
      edm:dataProvider ?provider .
  FILTER(isLiteral(?provider))
}
LIMIT 100
```

Listing 1.2: Objects associated with a given place.

```
\begin{lstlisting}[language=SPARQL, basicstyle=\ttfamily\small, breaklines=true]
PREFIX dc: <http://purl.org/dc/elements/1.1/>
PREFIX dcterms: <http://purl.org/dc/terms/>
SELECT ?cho ?title ?date WHERE {
```

³² <https://www.bildindex.de>

```

?cho dct:spatial <https://sws.geonames.org/2925533> ; #
  Frankfurt am Main
  dc:title ?title ;
  dc:date ?date .
FILTER(STRSTARTS(?date, "180"))
}
LIMIT 100

```

Listing 1.3: Source type strings alongside aligned mocho classes.

```

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dc: <http://purl.org/dc/elements/1.1/>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
SELECT ?mochoClass ?dctype (COUNT(DISTINCT ?cho) AS ?n) WHERE
{
  GRAPH <https://gemea.ise.fiz-karlsruhe.de/graph/mocho> {
    ?mocho rdf:type ?mochoClass ;
           owl:sameAs ?cho .
  }
  GRAPH <https://gemea.ise.fiz-karlsruhe.de/graph/ddbedm> {
    ?cho dc:type ?dctype .
  }
}
GROUP BY ?mochoClass ?dctype ORDER BY DESC(?n) LIMIT 20

```

5.2 Research Applications

GEMEA’s named graph structure, GND linking, provenance traces, and MCP interface support the following research use cases.

KG-grounded retrieval. The SPARQL endpoint and MCP interface support evaluation of KG-grounded RAG pipelines [13], entity-linked question answering [10], and natural language question answering over structured data [16].

Entity embedding and link prediction. The entity graph (agents, places, timespans, providers, `mocho:Work` groupings, and domain- and modality-specific classes emitted by the mocho alignment) provides a large-scale benchmark for entity embedding and link prediction methods [17], including LLM-enhanced and contrastive approaches. Named graph structure supports provenance-aware experiments that distinguish source triples from enrichment triples.

Explainability. Provenance-tracked named graphs and PROV-O[11] traces on GND-linked triples support evaluation of KG-grounded explanation methods, including tracing a retrieval result back to its source institution, enrichment pathway, and match confidence. The same structure supports evaluation of LLM-assisted enrichment provenance, where standard PROV-O traces are insufficient to capture model identity, confidence, and refinement type.

Metadata quality analysis. GEMEA exposes systematic encoding inconsistencies in source metadata that are invisible to record-level inspection but emerge through cross-graph queries. The following query identifies objects attributed to Goethe as `dc:creator` or `dc:contributor` in museum records, grouped by `dc:type` (see also Section 4):

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dc: <http://purl.org/dc/elements/1.1/>
PREFIX edm: <http://www.europeana.eu/schemas/edm/>
SELECT ?dctype (COUNT(DISTINCT ?cho) AS ?n)
WHERE {
  GRAPH <https://gemea.ise.fiz-karlsruhe.de/graph/ddbedm> {
    { ?cho dc:creator <http://d-nb.info/gnd/118540238> }
    UNION
    { ?cho dc:contributor <http://d-nb.info/gnd/118540238> }
    OPTIONAL { ?cho dc:type ?dctype }
    ?agg edm:aggregatedCHO ?cho ;
        edm:dataProvider ?inst .
    ?inst rdf:type <http://ddb.vocnet.org/sparte/sparte006> .
  }
}
GROUP BY ?dctype ORDER BY DESC(?n)
```

The query returns 120 records from the museum sector (`sparte006`) across `dc:type` values including prints (59), stage productions (30), medals (5), and minting dies (2). Within this set, 8 numismatic records (medals, minting dies, banknotes) show a clear misattribution: `dc:title` follows the pattern *Maker: Johann Wolfgang von Goethe* (e.g., “Bovy, Jean François Antoine: Johann Wolfgang von Goethe”), where Goethe is the depicted subject but was recorded as `dc:creator` rather than the actual maker. This encoding error, propagated from institutional source records through the DDB aggregator, is invisible without cross-institution graph queries.

Ontology evaluation. MOCHO-aligned subgraph provides a grounded evaluation target for ontology alignment methods. Because `graph/ddbedm` and `graph/mocho` encode the same objects under different schemas, the same query intent can be expressed against both graphs and compared on SPARQL execution time, result set size, and query complexity. Typed properties and WEMI-level distinctions in the aligned graph should reduce query verbosity and improve selectivity relative to the flat EDM representation; divergence on these metrics gives a measurable alignment quality signal without a manually curated gold standard.

5.3 Self-Hosting

The full GEMEA stack is deployable via Docker Compose, with setup scripts for multiple configurations (full stack, endpoint-only, MCP proxy) provided in the repository. Operators should expect at least 2 TB of disk space for the QLever index and source data.

6 Impact and Adoption

6.1 Access to the DDB Corpus

GEMEA is the first open, SPARQL-accessible knowledge graph over the DDB corpus, making the digitized collection queryable by SPARQL, dereferenceable as Linked Data, and accessible to AI agents via MCP.

6.2 FAIR Compliance

GEMEA satisfies the FAIR data principles [19]:

- **Findable.** The dataset is publicly available for download at <https://gemea.fiz-karlsruhe.de/downloads/>.
- **Accessible.** Data is retrievable via a public SPARQL endpoint and through the SHMARQL Linked Data browser.
- **Interoperable.** Standard RDF serializations (N-Triples, Turtle, JSON-LD), established vocabularies (EDM, RDA, FRBR, Dublin Core, GeoNames), and GND authority identifiers for agents and works are used throughout.
- **Reusable.** Source code, pipeline scripts, and MCP setup are released at <https://github.com/anntanp/gemea/> under the MIT licence. Individual object rights are as declared by the contributing institution (typically CC 1.0) and are queryable via `dcterms:rights` in the graph.

6.3 Target Communities

Digital humanities and library science. German CH research has lacked a machine-readable, graph-accessible corpus at this scale. GEMEA enables quantitative analysis previously infeasible without custom DDB API scraping: cross-sector co-occurrence patterns, temporal distribution of holdings, and geographic coverage.

Semantic Web and KG research. A public SPARQL endpoint, named graph provenance, MOCHO ontology alignment, and MCP agentic access make GEMEA a testbed for KG construction methods, ontology evaluation, entity linking, and LLM + KG research (Section 5).

Knowledge graph embedding research. GEMEA’s MOCHO-aligned representation of bibliographic titles opens a concrete empirical question that remains partially explored in the KGE-with-literals literature [9, 20, 18, 3]: whether entity descriptions yield better embeddings when modeled as a single rich literal or as multiple typed relations with shorter, semantically structured values. Existing approaches incorporate literals either by embedding them directly (e.g., LiteralE [9], DKRL [20], KEPLER [18]) or by transforming them into graph structure [3]. However, these methods treat the representation of literals as a modeling choice fixed at preprocessing time, and do not systematically compare alternative semantic decompositions of the same information. In particular, prior

work focuses on syntactic or structural transformations of literals (e.g., converting values into entities), rather than semantically grounded decompositions aligned with domain ontologies. GEMEA makes this comparison tractable: the MOCHO-aligned subgraph exposes both the decomposed RDA title structure and the source `dc:title` literals within the same graph, enabling controlled ablation studies at scale on a real-world cultural heritage corpus.

Cultural heritage institutions. GEMEA provides a self-hostable reference deployment that any cultural heritage aggregator can adapt: the JSON ingest + `mocho` + QLever stack, with SHMARQL as a Linked Data browser, MCP/MCPO integration to commercial AI assistants and other self-hosted setups (Open WebUI + Ollama). The full stack is documented, containerized, and deployable via Docker Compose (Section 7). The more than 500 institutions that have contributed data to the DDB (out of over 5,000 registered) can query their own holdings in graph form through GEMEA, enabling metadata introspection, coverage analysis, and targeted enrichment that feeds back into the aggregator.

7 Sustainability

Versioning. Per-item provenance is recorded at ingest time using the W3C PROV Ontology [11]: each `edm:ProvidedCHO` carries `prov:generatedAtTime`, `dcterms:hasVersion` (DDB revision id), and links to the source dataset and mapping-version agent derived from the DDB JSON metadata. Graph-level pipeline provenance — an Activity record per script run, capturing agent identity, version, timestamps, and an `inferenceMethod` flag (`Deterministic` / `Heuristic` / `NER`) — is planned as future work.

Update cadence. GEMEA is updated in sync with major DDB data releases.

Persistent URI. The dataset landing page at <https://gemea.ise.fiz-karlsruhe.de/> serves as the stable reference URI, with downloads at <https://gemea.ise.fiz-karlsruhe.de/downloads/> and the SHMARQL browser at <https://gemea.ise.fiz-karlsruhe.de/shmarql/>.

Open source. The GEMEA index is reproducible from the DDB data dump using the pipeline scripts in the repository.

Institutional support. GEMEA is developed at FIZ Karlsruhe – Leibniz Institute for Information Infrastructure in collaboration with the Karlsruhe Institute of Technology (AIFB). The resource is hosted on FIZ Karlsruhe infrastructure and maintained as part of the institute’s long-term research data services.

Dependency management. The MOCHO alignment layer depends on `mocho.owl`, which is under active development. Alignment rule changes upstream are tracked in the GEMEA repository and trigger a re-run of the affected named graph. Known scope boundaries in the current alignment (Section 4) are documented as versioned issue records.

Outbound URI persistence. As DDB does not operate under a formal persistent URI policy, objects withdrawn or re-harvested by contributing institutions will render a proportion of these links unreachable over time.

8 Conclusion

GEMEA provides the first unified graph and agent-based access to 26.8 million cultural heritage objects from the DDB corpus through a SPARQL endpoint, a Linked Data browser, and an MCP-based interface. By exposing the same large-scale CH KG through both SPARQL and agent-oriented interfaces, GEMEA enables direct integration of structured CH data into frontier NLP and LLM workflows. The resource is designed as a reusable infrastructure for researchers working on KG construction, ontology alignment, entity linking, and LLM-grounded retrieval over large-scale, heterogeneous CH data.

Resource Availability Statement. The GEMEA dataset is available for download at <https://gemea.ise.fiz-karlsruhe.de/downloads>. The SPARQL endpoint and Linked Data browser are served at <https://gemea.ise.fiz-karlsruhe.de/shmarql/>. Source code, pipeline scripts, and MCP setup are released at <https://github.com/anntanp/gemea/> under the MIT licence. Individual object rights are as declared by the contributing institution (typically CC 1.0) and are queryable via `dcterms:rights` in the graph.

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